

Expanding Double Brackets

AQA KS3 · Year 8 · Foundation

Name: _____ Class: _____ Date: _____

What you need to be able to do

Expand products of two binomials (double brackets) using the grid method or FOIL. Collect like terms to write a simplified quadratic expression.

Common mistakes to avoid

- Forgetting the last term: you must multiply the constant in the first bracket by the constant in the second.
- Sign errors with negatives: remember that a negative times a negative gives a positive.
- Writing $x + x$ instead of $2x$ for the middle term when collecting like terms.
- Forgetting that $x \times x = x^2$, not $2x$.

Method steps

1. Step 1: Draw a grid with the terms of the first bracket down the left and the terms of the second bracket along the top.

2. Step 2: Fill in each cell by multiplying the row header by the column header.
3. Step 3: Write out all four products in a line.
4. Step 4: Collect like terms (the two middle terms will usually be alike).
5. Step 5: Check by substituting a simple value such as $x = 10$.

Worked example

Expand and simplify: $(x + 2)(x + 3)$

Step 1 — Draw the grid: columns are x and $+3$, rows are x and $+2$.

Step 2 — Fill in: $x \cdot x = x^2$, $x \cdot 3 = 3x$, $2 \cdot x = 2x$, $2 \cdot 3 = 6$

Step 3 — Write out: $x^2 + 3x + 2x + 6$

Step 4 — Collect like terms: $3x + 2x = 5x$

Answer: $x^2 + 5x + 6$

Check ($x = 10$): $(12)(13) = 156$ and $100 + 50 + 6 = 156$

Expand and simplify.

$$(x + 1)(x + 2)$$

How to start

1. Draw a 2×2 grid. Put x and $+1$ down the left side, x and $+2$ along the top.
2. Fill in each cell: $x \cdot x$, $x \cdot 2$, $1 \cdot x$, $1 \cdot 2$.
3. Write all four terms, then collect the like terms together.

Answer: _____

Remember the method:

- 1 Draw the grid 2 Multiply each pair 3 Write all four products
4 Collect like terms 5 Check by substituting

Expand and simplify.

$$(x + 3)(x + 4)$$

How to start

1. Draw a 2×2 grid. Put x and $+3$ down the left side, x and $+4$ along the top.
2. Fill in each cell by multiplying the row term by the column term.
3. Add up the four products and collect like terms.

Answer: _____

Remember the method:

- 1 Draw the grid 2 Multiply each pair 3 Write all four products
4 Collect like terms 5 Check by substituting

Expand and simplify.

$$(x + 5)(x - 2)$$

Answer: _____

Remember the method:

- 1 Draw the grid 2 Multiply each pair 3 Write all four products
4 Collect like terms 5 Check by substituting

Expand and simplify.

$$(x - 3)(x + 4)$$

Answer: _____

Remember the method:

- 1 Draw the grid 2 Multiply each pair 3 Write all four products
4 Collect like terms 5 Check by substituting

Expand and simplify.

$$(x - 2)(x - 5)$$

Answer: _____

Remember the method:

- 1 Draw the grid 2 Multiply each pair 3 Write all four products
4 Collect like terms 5 Check by substituting

Expand and simplify.

$$(2x + 1)(x + 3)$$

Answer: _____

Remember the method:

- 1 Draw the grid 2 Multiply each pair 3 Write all four products
4 Collect like terms 5 Check by substituting

Expand and simplify.

$$(3x - 2)(x + 4)$$

Answer: _____

Remember the method:

- 1 Draw the grid 2 Multiply each pair 3 Write all four products
4 Collect like terms 5 Check by substituting

Expand and simplify.

$$(2x + 3)(3x - 1)$$

Answer: _____

Remember the method:

- 1 Draw the grid 2 Multiply each pair 3 Write all four products
4 Collect like terms 5 Check by substituting

Self-reflection

How confident do you feel with this topic now? (tick one)

☐ Not yet ☐ Getting there ☐ Confident ☐ Really confident

Which question did you find the hardest, and why?

One thing I now understand about expanding double brackets:

One thing I will practise before the next lesson:

Teacher key

Answers

1. Q1: $x^2 + 3x + 2$ $[(x+1)(x+2): x^2 + 2x + x + 2 = x^2 + 3x + 2]$

2. Q2: $x^2 + 7x + 12$ $[(x+3)(x+4): x^2 + 4x + 3x + 12 = x^2 + 7x + 12]$

3. Q3: $x^2 + 3x - 10$ $[(x+5)(x-2): x^2 - 2x + 5x - 10 = x^2 + 3x - 10]$

4. Q4: $x^2 + x - 12$ $[(x-3)(x+4): x^2 + 4x - 3x - 12 = x^2 + x - 12]$

5. Q5: $x^2 - 7x + 10$ $[(x-2)(x-5): x^2 - 5x - 2x + 10 = x^2 - 7x + 10]$

6. Q6: $2x^2 + 7x + 3$ $[(2x+1)(x+3): 2x^2 + 6x + x + 3 = 2x^2 + 7x + 3]$

7. Q7: $3x^2 + 10x - 8$ $[(3x-2)(x+4): 3x^2 + 12x - 2x - 8 = 3x^2 + 10x - 8]$

8. Q8: $6x^2 + 7x - 3$ $[(2x+3)(3x-1): 6x^2 - 2x + 9x - 3 = 6x^2 + 7x - 3]$